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(54) **LIGHTING DEVICE**

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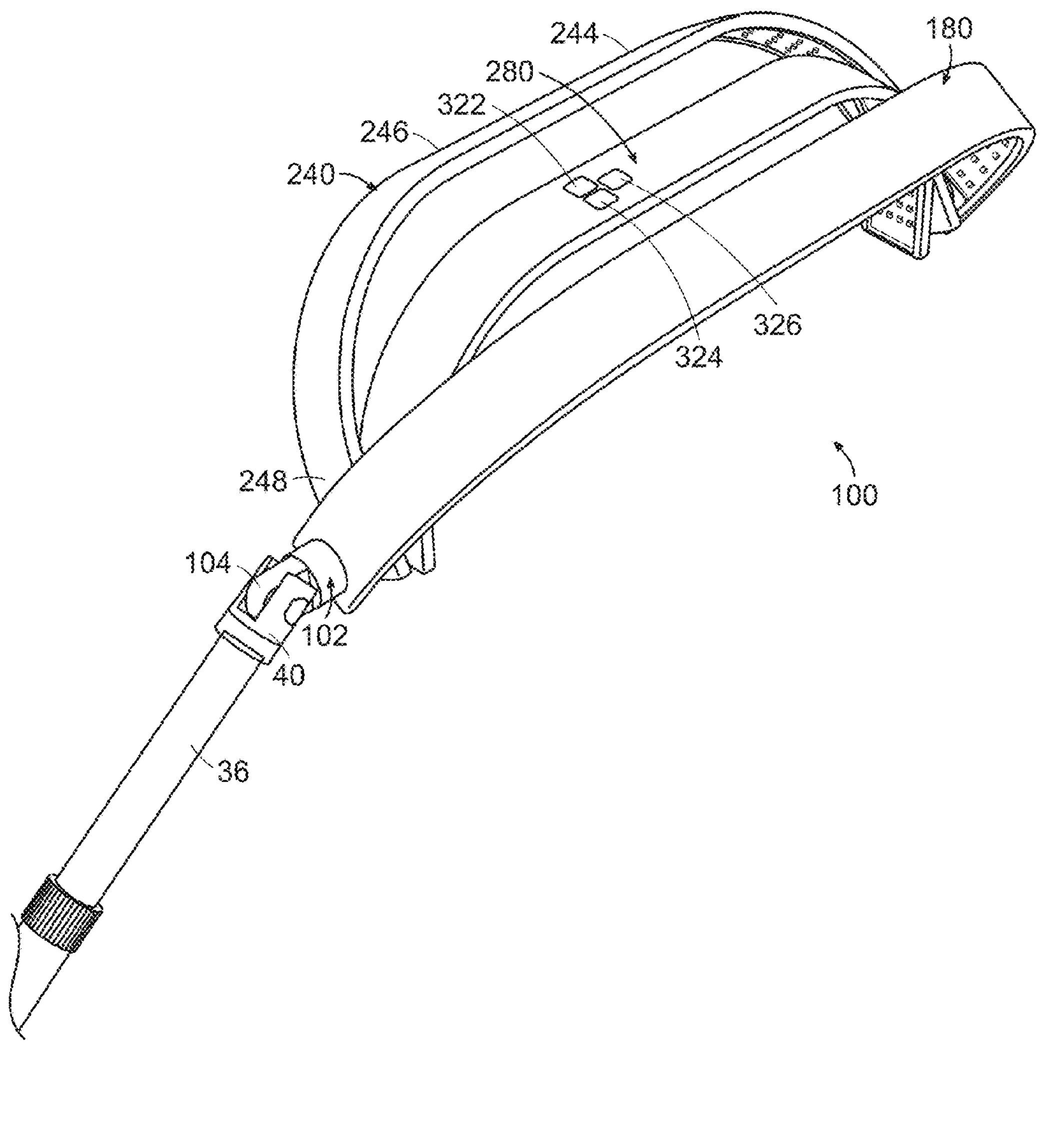
(57) **ABSTRACT**

A lighting device for use by a person comprising an inner bracket and an outer bracket. The device further comprises a first light rotatably engaged the inner bracket and the outer bracket. The device further comprises a second light rotatably engaged with the inner bracket and the outer bracket. The second light is disposed below the first light. The device further comprises a third light engaged with the inner bracket and the outer bracket. The third light is disposed below the second light.



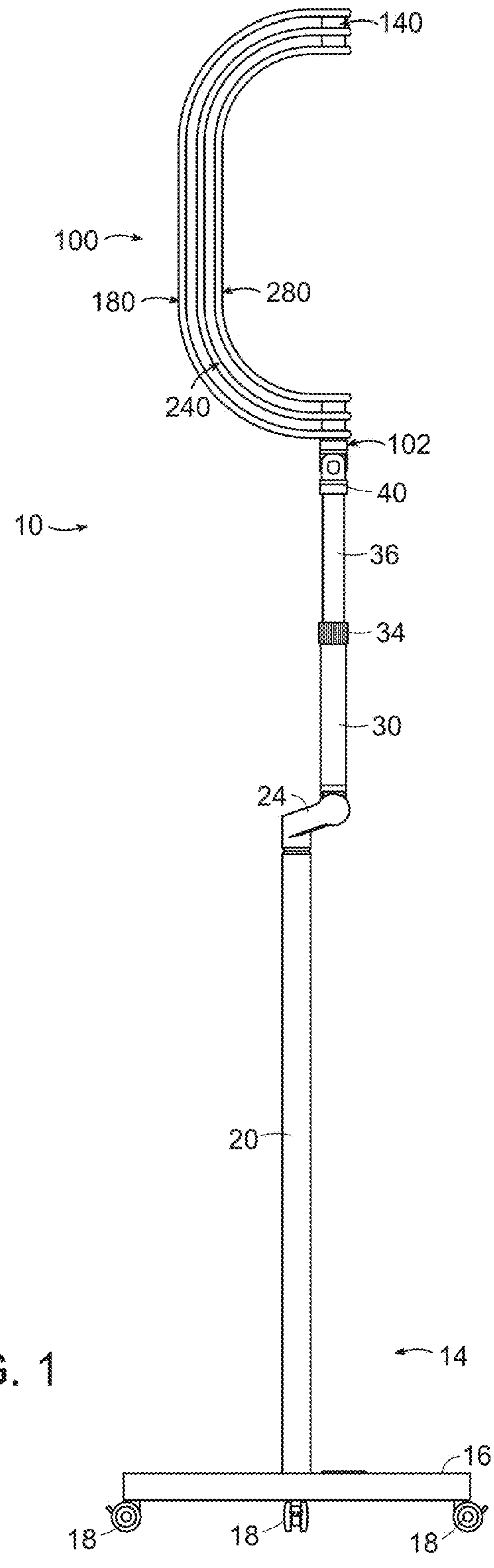
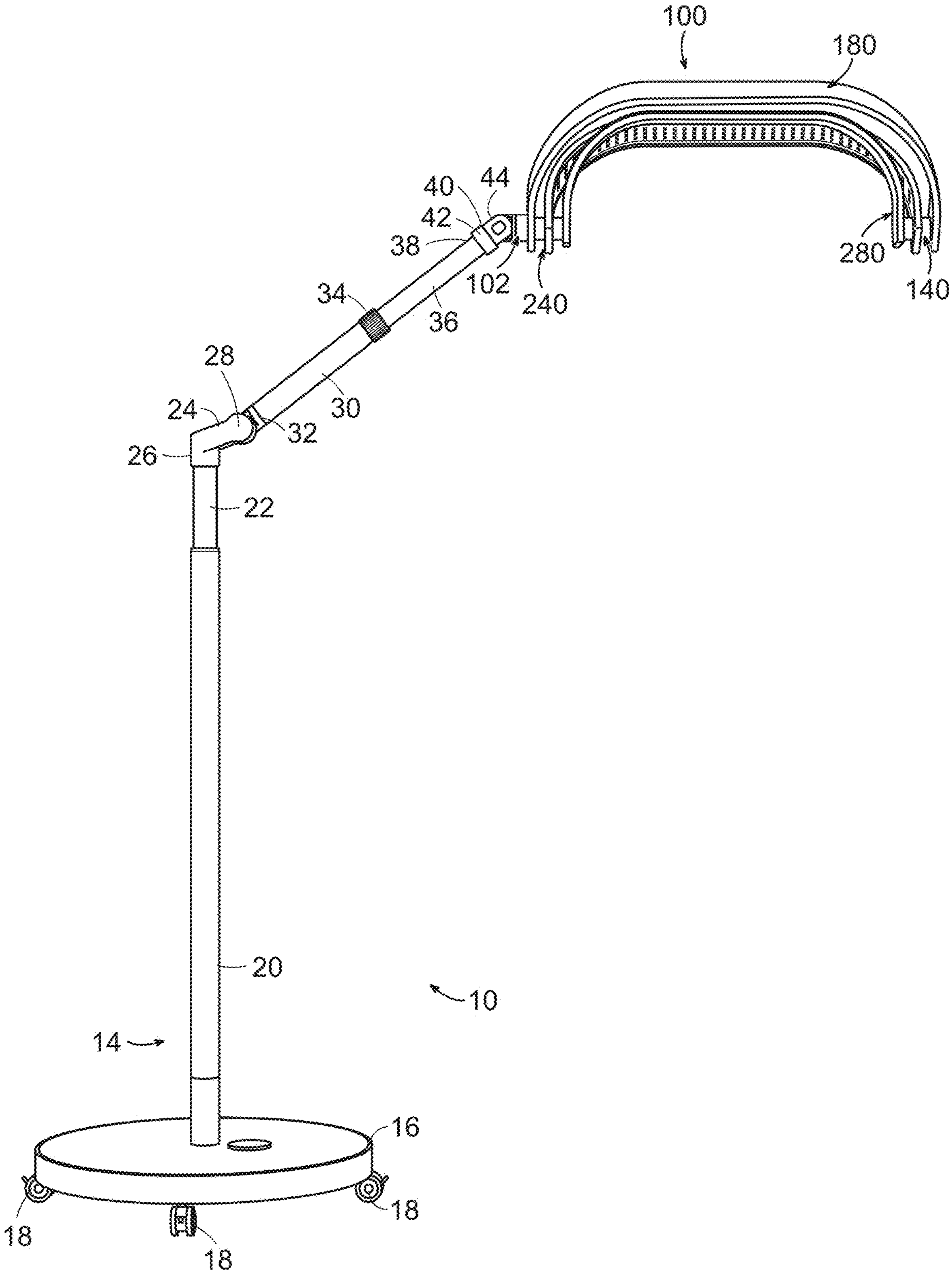


FIG. 1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180
40
44
42
38
102
240
280
140
34
36
28
30
24
32
26
22
20
10
14
16
18
18
18

FIG. 2

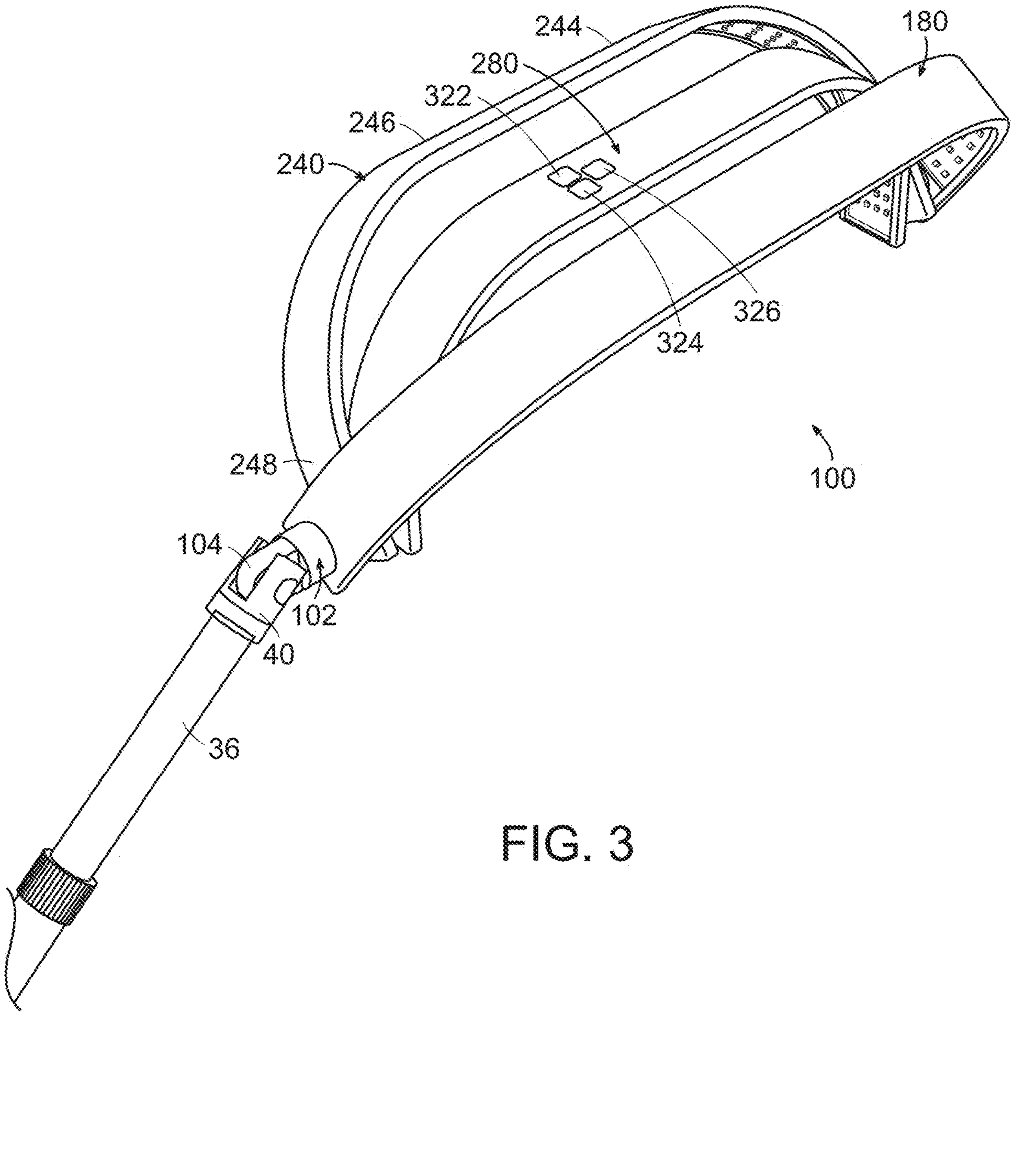
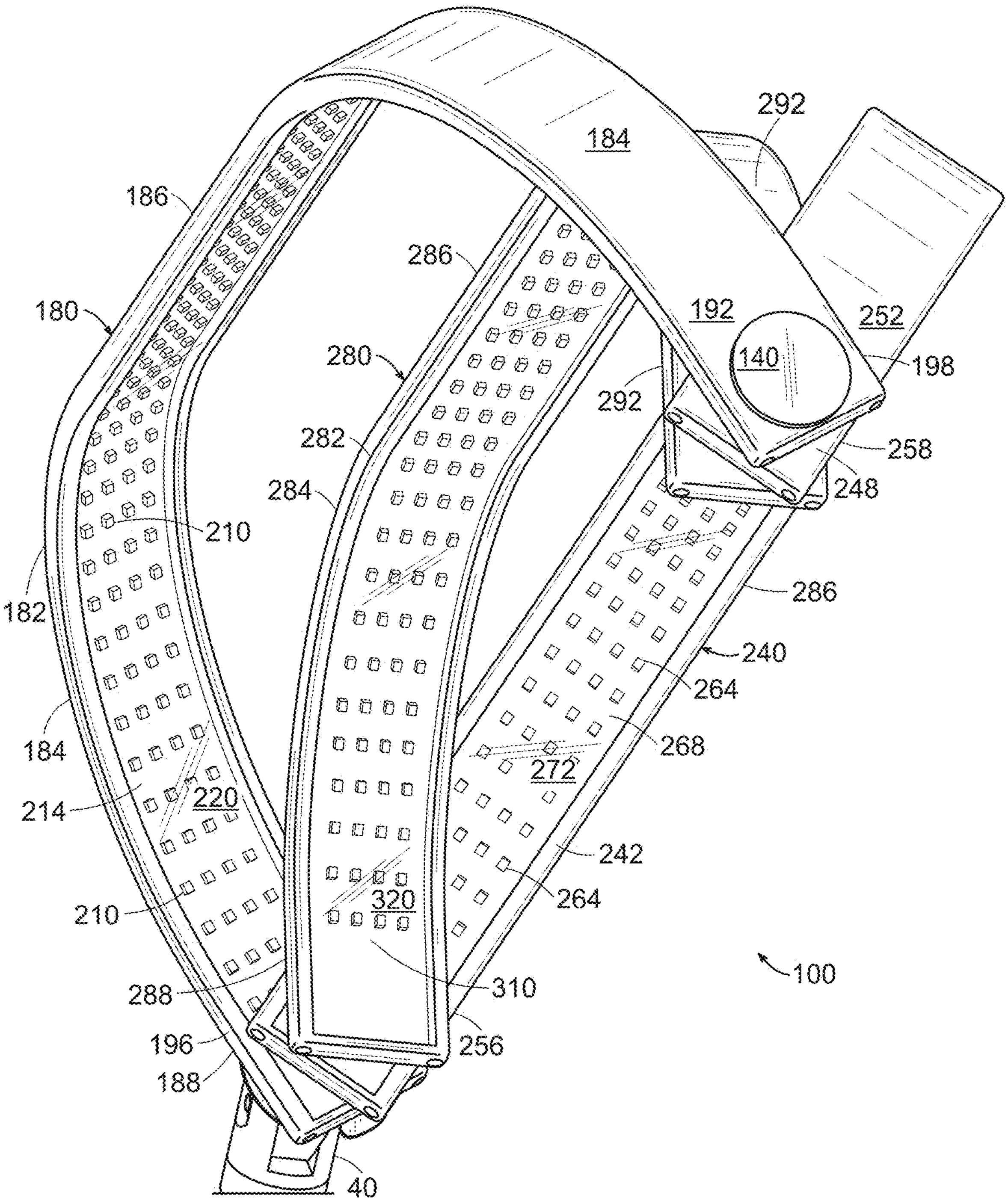


FIG. 3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184
252
186
286
192
180
280
140
198
292
282
258
248
284
210
286
240
182
264
268
184
272
214
220
242
264
210
320
100
288
310
196
256
188
40

FIG. 4

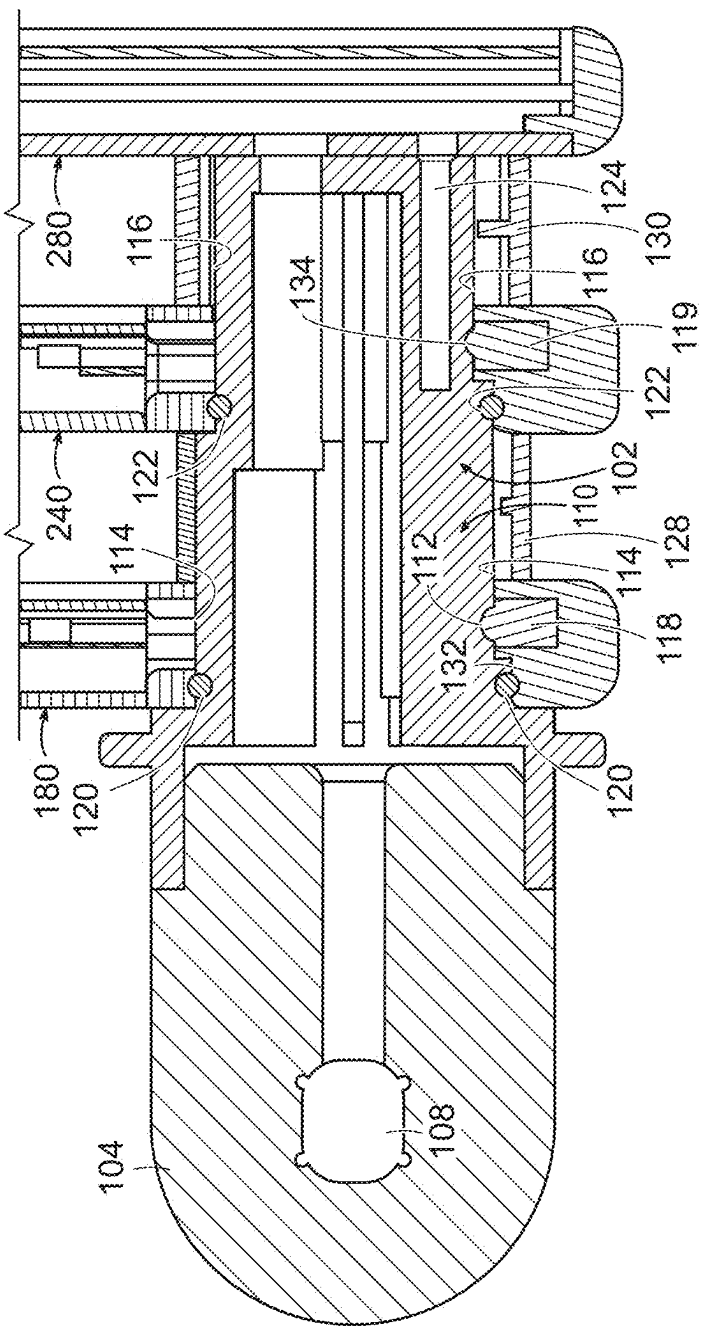
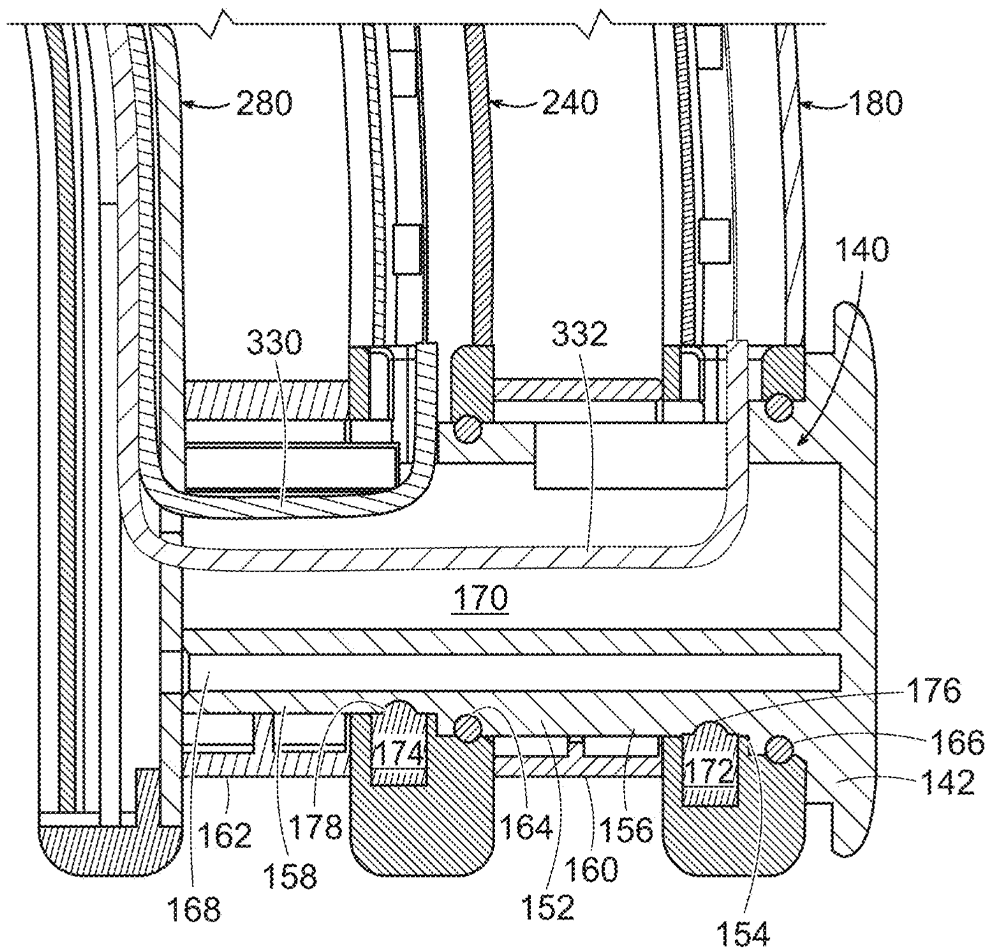

280
116
134
124
130
116
119
122
122
102
110
240
114
112
128
114
118
132
180
120
120
104
108

FIG. 5


280
240
180
140
330
332
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176
166
142
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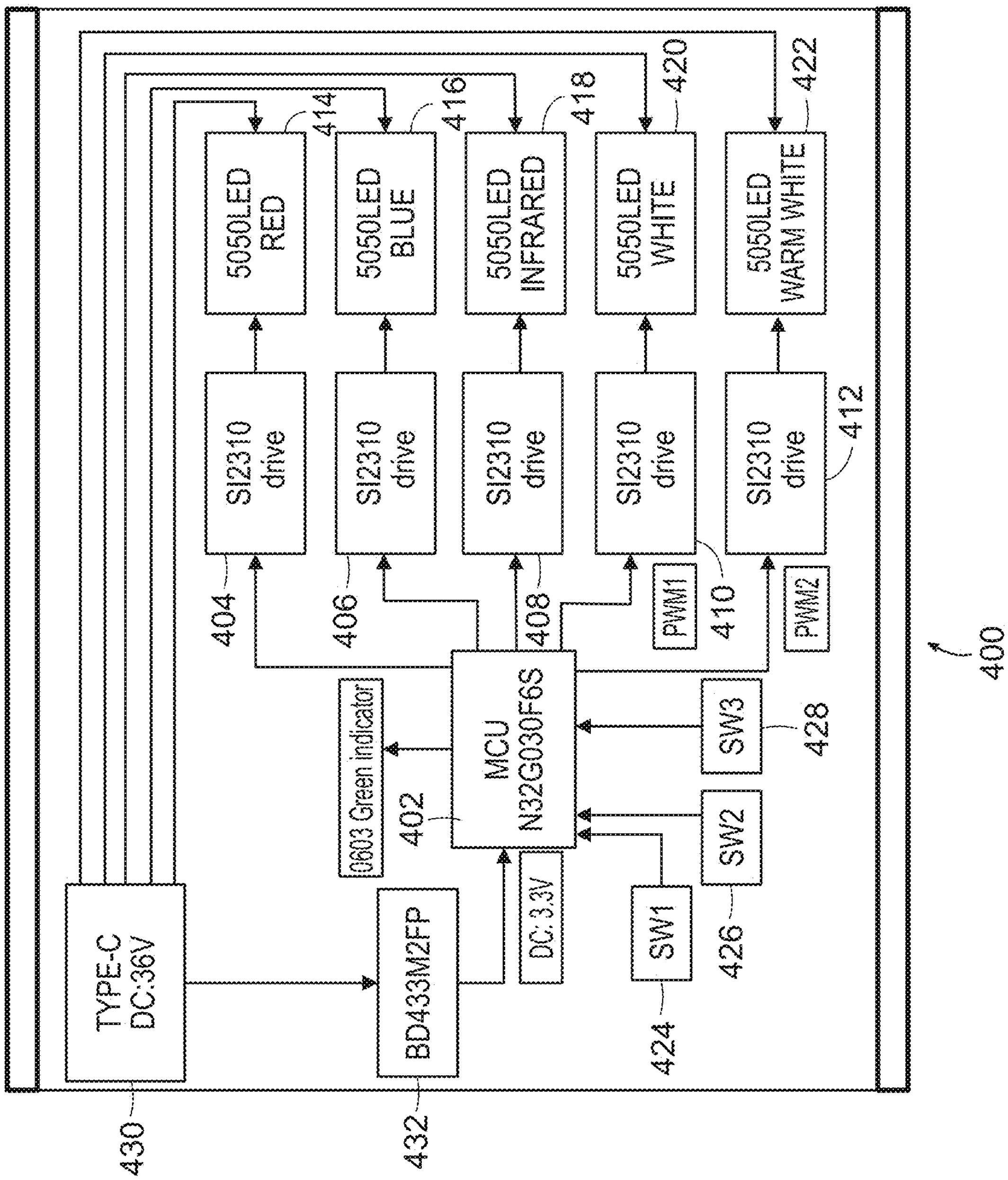
172
162
178
164
156
158
160
168
152
154

FIG. 6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BD433M2FP
DC: 3.3V
MCU
N32G030F6S
0603 Green indicator
SW1
SW2
SW3
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drive
SI2310
drive
SI2310
drive
SI2310
drive
SI2310
drive
PWM1
PWM2
5050LED
RED
5050LED
BLUE
5050LED
INFRARED
5050LED
WHITE
5050LED
WARM WHITE
400
402
404
406
408
410
412
414
416
418
420
422
424
426
428
430
432

FIG. 7

LIGHTING DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to and the benefit of U.S. Provisional Application Ser. No. 63/559,761 filed on Feb. 29, 2024; U.S. Provisional Application Ser. No. 63/676,418 filed on Jul. 28, 2024; U.S. Provisional Application Ser. No. 63/678,858 filed on Aug. 2, 2024; U.S. Provisional Application Ser. No. 63/682,314 filed on Aug. 12, 2024; and U.S. Provisional Application Ser. No. 63/686,607 filed on Aug. 23, 2024.

BACKGROUND OF THE INVENTION

[0002] Light therapy employing light emitting diodes have recently been used to trat facial conditions such as acne and other conditions. There is a continuing need for ergonomic devices to deliver such light therapy.

SUMMARY OF THE INVENTION

[0003] A light device for use by a person comprising an inner bracket and an outer bracket. The device further comprises a first light rotatably engaged the inner bracket and the outer bracket. The device further comprises a second light rotatably engaged with the inner bracket and the outer bracket. The second light is disposed below the first light. The device further comprises a third light engaged with the inner bracket and the outer bracket. The third light is disposed below the second light. The first light, second light, and third light each comprise a plurality of light sources that in the preferred embodiment are light emitting diodes having different types of colors to allow for different types of light treatment therapy.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The following description of the invention will be more fully understood with reference to the accompanying drawings in which:

[0005] FIG. **1** is a perspective view of a lighting device according to the present invention mounted to a stand in a first position;

[0006] FIG. **2** is a perspective view of the lighting device mounted to the stand in a second position;

[0007] FIG. **3** is a top perspective view of a light unit of the lighting device in a position selected by a person shown comprising a first light and a second light rotatably engaged with an inner bracket and an outer bracket of the light unit, and a third light engaged with the inner bracket and outer bracket of the light unit;

[0008] FIG. **4** is a bottom perspective view of the first light and second light rotatably engaged with the inner bracket and outer bracket of the light unit and the third light engaged with the inner bracket and outer bracket of the light unit;

[0009] FIG. **5** is a cross-section view of the first light and second light rotatably engaged with the inner bracket of the light unit and the third light engaged with the inner bracket of the light unit;

[0010] FIG. **6** is a cross-section view of the first light and the second light rotatably engaged with the outer bracket of the light unit and the third light engaged with the outer bracket of the light unit; and

[0011] FIG. **7** is a high level block schematic of an electronic circuit for controlling the light emitting diodes of the first light, the second light, and the third light.

DESCRIPTION OF THE INVENTION

[0012] Referring to FIGS. **1-2**, a lighting device **10** according to the present invention is shown for use by a person or practitioner such as a cosmetologist to provide a plurality of LED lights for a variety of purposes and applications during cosmetic procedures such as LED light therapy with different colors for treating various facial and/or other skin ailments. Lighting device **10** generally comprises a light unit **100** rotatably engaged with a stand **14**.

[0013] With continued reference to FIGS. **1** and **2**, stand **14** comprises a base **16** and a plurality of wheels **18** mounted to the bottom of base **16** to allow lighting device **10** to be easily moved. Stand **14** further comprises a tubular pole **20** attached to base **16** by conventional means and a tubular pole **22** telescopically disposed and moveable with tubular pole **20** by conventional means. Stand **14** further comprises a bracket **24** comprising a lower end **26** engaged with tubular pole **22** and an upper end **28**. Stand **14** further comprises a tubular pole **30** comprising a lower end **32** rotatably engaged with upper end **28** of bracket **24**. Tubular pole **30** further comprises a collar **34** for locking a tubular pole **36** in a position selected by the person. Tubular pole **36** is telescopically engaged within tubular pole **30** and can be locked in place by tightening collar **34**. Tubular pole **36** further comprises an upper end **38** that is securely engaged with a mounting bracket **40**. Mounting bracket **40** comprises a lower end **42** engaged with upper end **38** of tubular pole **36** and an upper end **44** that is rotatably engaged with light unit **100**. Stand **14** and all of its components are well known in the art and widely available.

[0014] With continued reference to FIGS. **1** and **2**, light unit **100** comprises an inner bracket **102** and an outer bracket **140**. Light unit **100** further comprises a first light **180** rotatably engaged with inner bracket **102** and outer bracket **140**. Light unit **100** further comprises a second light **240** rotatably engaged with inner bracket **102** and outer bracket **140**. Light unit **100** further comprises a third light **280** engaged with inner bracket **102** and outer bracket **140**. Second light **240** is disposed below first light **180**, and third light **280** is disposed below second light **240**. First light **180** and second light **240** can be rotated outward from third light **280** to form a partial or full dome shape as shown in FIG. **2**. First light **180**, second light **240**, and third light **280** can be placed in many positions such as a vertical overlapping position as shown in FIG. **1**. First light **180** and second light **240** can be rotated to many different positions other than as shown in FIGS. **1** and **2**.

[0015] Referring to FIGS. **3** and **4**, first light **180** comprises a housing **182** comprising a rear wall comprising a median portion **186**, a left end **188** rotatably engaged with inner bracket **102**, and a right end **192** rotatably engaged with outer bracket **140**. Left end **188** is curved downward and inward from median portion **186**. Right end **192** is curved downward and inward from median portion **186** towards left end **188**. Housing **182** further comprises a circular shaped opening **190** (not shown) extending completely thru left end **188** to allow left end **188** of housing **182** to rotatably engage with inner bracket **102**. Housing **182** further comprises a circular shaped opening **194** (not shown) extending completely thru right end **192** to allow right end

192 of housing 182 to rotatably engage with outer bracket 140. First light 180 further comprises a plurality of light sources 210 disposed within housing 182 substantially extending from left end 188 to right end 192. In the embodiment shown, light sources 210 are light emitting diodes. First light 180 further comprises a printed circuit board 214 disposed within housing 182 substantially extending from left end 188 to right end 192 for mounting and electrically connecting light sources 210 by conventional means. First light 180 further comprises a transparent cover 220 attached to housing 182 by conventional means such as grooves disposed in the side walls of housing 182. Light sources 210 are disposed below transparent cover 220 such that light from light sources 210 pass thru transparent cover 220. Transparent cover 220 can take other forms such as frosted.

[0016] With continued reference to FIGS. 3 and 4, second light 240 comprises a housing 242 comprising a rear wall 244 comprising a median portion 246, a left end 248 rotatably engaged with inner bracket 102 and a right end 252 rotatably engaged with outer bracket 140. Left end portion 248 is curved downward and inward from median portion 246. Right end portion 252 is curved downward and inward from median portion 246 towards left end portion 248. Housing 182 further comprises a circular shaped opening 250 (not shown) extending completely thru left end 248 to allow left end 248 of housing 242 to rotatably engage with inner bracket 102. Housing 182 further comprises a circular shaped opening 254 (not shown) extending completely thru right end 252 to allow right end 252 of housing 182 to rotatably engage with outer bracket 140. Second light 240 further comprises a plurality of light sources 264 disposed within housing 242 substantially extending from left end 248 to right end 252. In the embodiment shown, light sources 264 are light emitting diodes. Second curved light 240 further comprises a printed circuit board 268 disposed within housing 242 substantially extending from left end 248 to right end 252 for mounting and electrically connecting light sources 264 by conventional means. Second light 240 further comprises a transparent cover 272 attached to housing 242 by conventional means such as grooves disposed in the side walls of housing 242. Light sources 264 are disposed below transparent cover 272 such that light from light sources 264 passes thru transparent cover 272. Transparent cover 272 may take other forms such as frosted.

[0017] With continued reference to FIGS. 3 and 4, third light 280 comprises a housing 282 comprising a rear wall 284 comprising median portion 286, a left end 288 rotatably engaged with inner bracket 102 and a right end 292 rotatably engaged with outer bracket 140. Left end portion 288 of rear wall 284 is secured or fixed to inner bracket 102 by a plurality of screws (not shown). In the embodiment shown, third light 280 does not rotate relative to inner bracket 102. In other embodiments, third curved light 280 may rotate relative to inner bracket 102. Left end 288 is curved downward and inward from median portion 286. Right end 292 is curved downward and inward from median portion 286 towards left end portion 288. Third light 280 further comprises a plurality of light sources 300 disposed within housing 282 substantially extending from left end 288 to right end 292. In the embodiment shown, light sources 300 are light emitting diodes. Third light 280 further comprises a printed circuit board 310 disposed within housing 282 substantially extending from left end 288 to right end 292 for mounting and electrically connecting light sources 300 by conventional means. Third light 280 further comprises a transparent cover 320 attached to housing 242 by conventional means such as grooves disposed in the side walls of housing 282. Light sources 300 are disposed below transparent cover 320 such that light from light sources 300 passes thru transparent cover 272. Transparent cover 3202 may take other forms such as frosted. As shown in FIG. 3, as will be described more fully herein, third light 280 further comprises a series of buttons 322, 324, and 326 disposed on median portion 286 of rear wall 284 of housing 282 that when depressed control the on/off state, color, and/or brightness of light sources 210 of first light 180, light sources 264 of second light 240, and/or light sources 300 of third light 280.

[0018] Referring to FIG. 5, a cross-section view shows first light 180 and second light 240 rotatably engaged with inner bracket 102 and third light 280 engaged with inner bracket 102. Inner bracket 102 comprises a left part 104 adapted to rotatably engage with bracket 40 of stand 12. Left part 104 comprises a channel 106 for wiring and an opening 108 for rotatably mounting with bracket 40. Inner bracket 102 further comprises a right part 110 adapted to rotatably receive first light 180 and second light 240 and engage with third light 280. Right part 110 comprises an outermost annular step or surface 112, a middle annular step or surface 114, and an innermost annular step or surface 116. Opening 190 of first light 180 is disposed about outermost annular surface 116 and middle annular surface 114 of right part 110. A spacer 128 is provided adjacent first light 180. Opening 250 of second light 240 is disposed about middle annular surface 114 and innermost annular surface 116 of right part 110. Right part 140 further comprises a hole 124 so that third light 280 can be screwed to right part 110 by a screw (not shown). Although one (1) hole is shown, three (3) holes and screws are employed so that third light 280 is fixed to right part 152. A spacer 130 is provided adjacent second light 240 and third light 280. An oaring 120 is disposed between outermost annular surface 112 of right part 110 and left end 196 of housing 182 of first light 180 to reduce friction during rotation. An O-rings 122 is disposed between middle annular surface 114 of right part 110 and left end 246 of housing 242 of second light 240 to reduce friction during rotation. A retractable pin 118 is provided in left end 196 of housing 182 of first curved light 180 that removably engages and/or locks with an indentation 132 of right part 110 of inner bracket 102 to secure first light 180 in a selected position. Multiple indentations may be provided so first light 180 can be rotated to different positions. A retractable pin 119 is provided in left end 256 of housing 242 of second light 240 that removably engages and/or locks with an indentation 134 of inner bracket 102 to secure second light 240 is a selected position. Multiple indentations may be provided so second light 180 can be rotated to different positions. Pin 118 and pin 119 are well known in the art and widely available.

[0019] Referring to FIG. 6, a cross-section view shows first light 180 and second light 240 rotatably engaged with outer bracket 140 and third light 280 engaged with outer bracket 140. Outer bracket 140 comprises a cap or flange 142 and a left part 152 adapted to rotatably engage with first light 180 and second light 240 and engage with third light 280. Left part 152 comprises a channel 170 for passing of wiring from third light 280 to first light 180 and second light 240. Wires 332 are provided to electrically connect an electronic circuit **400** (FIG. **4**—to be described) of third light **280** with light sources **210** of first light **180**. Wires **330** are provided to electrically connect electronic circuit **400** of third light **280** with light sources **264** of second light **240**. Left part **152** comprises an outermost annular step or surface **154**, a middle annular step or surface **156**, and an innermost annular step or surface **158**. Opening **194** of first light **180** is inserted and disposed about outmost annular surface **154** and middle annular surface **156** of left part **152**. A spacer **160** is provided adjacent first light **180**. Opening **254** of second light **240** is inserted and disposed about middle annular surface **156** and innermost annular surface **158** of left part **152**. Left part **152** further comprises a hole **168** so that third light **280** can be screwed to left part **152** by a screw (not shown). Although one (1) hole is shown, three (3) holes and screws are employed so that third light **280** is fixed to left part **152**. A spacer **162** is provided adjacent second light **240** and third light **280**. An o-ring **166** is disposed between innermost annular surface **154** of left part **152** and right end **198** of housing **182** of first light **180** to reduce friction during rotation. An o-ring **164** is disposed between middle annular surface **156** of left part **152** and right end **258** of housing **242** of second light **240** to reduce friction during rotation. A pin **172** is provided in right end **198** of housing **182** of first curved light **180** that removably engages and/or locks with an indentation **176** of left part **152** of outer bracket **140** to secure first light **180** is a selected position. Multiple indentations may be provided so first light **180** can be rotated to different positions. A pin **174** is provided in right end **258** of housing **242** of second light **240** that removably engages and/or locks with an indentation **178** of outer bracket **140** to secure second light **240** is a selected orientation. Multiple indentations may be provided so second light **240** can be rotated to different positions. Pins **172** and **174** are well known in the art and widely available.

[0020] Third light **280** further comprises an electronic circuit **400** (FIG. **4**) connected to printed circuit board **310**, or another printed circuit board, to power and control operation of light sources **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third light **280** such as light color and/or temperature. Light sources **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third light **280** may have the same or different color and/or temperatures depending upon the desired application and/or treatment. Electronic circuit **400** of third light **280** is electrically connected to printed circuit board **214** of first light **180** and printed circuit board **268** of second light **240** by conventional wiring as previously described. Electronic circuits comprising microcontrollers to control the color and/or temperature of a single set or multiple arrays of Light emitting diodes are well known in the art. By way of example only, an electronic circuit for controlling light emitting diodes is described in U.S. Pat. No. 9,648,217 that is hereby incorporated into this specification in its entirety.

[0021] Referring to FIG. **7**, where a preferred embodiment of electronic circuit **400** is generally shown. Electronic circuit **400** comprises a microcontroller **402** that may be programmed by well known means in the art to activate and/or control light emitting diodes **210** of first curved light **180**, light emitting diodes **264** of second curved light **240** and light emitting diodes **300** of third curved light **280**. Microcontroller **402** is connected to a LED driver **404** that is connected to a red LED array **414**. Microcontroller **402** is connected to a LED driver **406** that is connected to a blue LED array **416**. Microcontroller **402** is connected to a LED driver **408** that is connected to an infrared LED array **418**. Microcontroller **402** is connected to a LED driver **410** that is connected to a white LED array **420**. Microcontroller **402** is connected to a LED driver **412** that is connected to a warm white LED array **420**. Each of light sources **210** of first light **180**, light sources **264** of second light **240** and light sources **300** of third light **280** are made up of red LED array **414**, blue LED array **416**, infrared LED array **418**, white LED array **420**, and warm white LED array **420** and electrically connected in the same manner. Electronic circuit **400** further comprises a first switch **424** connected to microcontroller **402**. First switch **424** is activated by depression of button **322** that sends a signal to microcontroller **402** to turn on and off the light sources of first light **180**, second light **240**, and third light **280**. Electronic circuit **400** further comprises a second switch **426** connected to microcontroller **402**. Second switch **426** is activated by depression of button **324** that sends a signal to microcontroller **402** to change the color mode of the light sources of first light **180**, second light **240**, and third light **280** in any programmed color. Electronic circuit **400** further comprises a third switch **428** connected to microcontroller **402**. Third switch **428** is activated by depression of button **326** that sends a signal to microcontroller **402** to adjust the brightness of the light sources of first light **180**, second light **240**, and third light **280** in any predetermined level. Electronic circuit **400** further comprises a USB Type C connector **430** connected to microcontroller **402** by a voltage regulator **432** that drops the 36V input to 3.3V for microcontroller **402**. Device **10** further comprises a conventional electrical cord (not shown) having a conventional AC/DC power supply (not shown) that a person can plug into a common electrical wall outlet and an USB connector that connects with USB connector **430** disposed on the bottom pole of stand **14** or many other places on stand **14** or light unit **100**. Conventional wiring is provided thru the poles and brackets to connect USB connector **430** with and provide power to printed circuit board **310** of third curved light **280**. Such electrical cords having a power supply are well known in the art and widely available as are USB connectors.

[0022] With reference to FIG. **3** and FIG. **7**, button **326** is used to turn on and off lighting device **10**. Depression and hold for two (2) seconds of button **326** causes a signal from switch **424** to be sent to microcontroller **402** of electronic control circuit **400** to turn on or off light sources **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third light **280**.

[0023] With continued reference to FIG. **3** and FIG. **7**, button **322** is used to select the color mode of light sources **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third light **280**. The default mode at startup (power turned on) are the colors red and infrared by activation of light emitting diodes **414** and **418** of the light sources **210**, **264**, and **300**. Pressing button **322** causes a signal from switch **426** to be sent to a microcontroller **402** of electronic control circuit **400** of third light **280** to activate the next color mode, namely, blue light emitting diodes **416** of light sources diodes **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third light **280**. Pressing button **322** again causes microcontroller **402** of electronic control circuit **400** of third light **280** to activate the next color mode, namely, "warm white light" light emitting diodes of **422** of light sources **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third curved light **280**. Pressing button **322** again causes microcontroller **402** of electronic control circuit **400** of third light **280** to activate the next color mode, namely, "cold white light" light emitting diodes of **420** of light sources **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third light **280**. Cycle thereafter.

[0024] With continued reference to FIG. **3** and FIG. **7**, pressing of button **324** adjusts the brightness of light sources **210** of first light **180**, light sources **264** of second light **240**, and light sources **300** of third light **280**. Different brightness are used from low to high cycle.

[0025] In other embodiments, any presently or futurely developed light source having therapeutic properties may be used instead of a light emitting diode light source. By way of further example only, stand **14** may be replaced with any other type of mounting means, such as table mounted means, with relocation of USB connector **430**.

[0026] The foregoing description is intended primarily for purposes of illustration. This invention may be embodied in other forms or carried out in other ways without departing from the spirit or scope of the invention. Modifications and variations still falling within the spirit or scope of the invention will be readily apparent to those of skill in the art.

What is claimed:

1. A device for use by a person comprising:

an inner bracket;

an outer bracket;

a first light rotatably engaged with said inner bracket and said outer bracket;

a second light rotatably engaged with said inner bracket and said outer bracket; said second light being disposed below said first light; and

a third light engaged with said inner bracket and said outer bracket; said third light being disposed below said second light.

2. The device of claim **1**, wherein said first light comprises a housing; said housing comprises a rear wall comprising a median portion, a first end, a second end, a first opening and a second opening thru said first end and said second end, respectively; said opening of said first end of said first light being rotatably engaged with said inner bracket; said opening of said second end of said first light being rotatably engaged with said outer bracket.

3. The device of claim **2**, wherein said second light comprises a housing; said housing comprises a rear wall comprising a median portion, a first end, a second end, and a first opening and a second opening thru said first end and said second end, respectively; said opening of said first end of said second light being rotatably engaged with said inner bracket; said opening of said second end of said second light being rotatably engaged with said outer bracket.

4. The device of claim **3**, wherein said third light comprises a housing; said housing comprises a rear wall comprising a median portion, a first end, and a second end; said first end of said third light is securely engaged with said inner bracket; and second end of said third light is securely engaged with said outer bracket.

5. The device of claim **4**, further comprising a first annular spacer disposed about said inner bracket and disposed between said first end of said first light and said first end of said second light.

6. The device of claim **5**, further comprising a second annular spacer disposed about said inner bracket and disposed between said first end of said second light and said first end of said third light.

7. The device of claim **6**, wherein said inner bracket comprises an outermost annular step engaged with said first light.

8. The device of claim **7**, wherein said inner bracket further comprises a middle annular step engaged with said second light.

9. The device of claim **1**, wherein said left end of said rear wall of said first light is curved downward and inward from said median portion of said rear wall of said first light; said right end of said rear wall of said first light is curved downward and inward from said median portion of said rear wall of said first light towards said left end portion of said rear wall of said first light.

10. The device of claim **9**, wherein said left end of said rear wall of said second light is curved downward and inward from said median portion of said rear wall of said first light; said right end of said rear wall of said first light is curved inward from said median portion of said rear wall of said first light towards said left end of said rear wall of said first light.

11. The device of claim **1**, wherein said first light comprises a plurality of light sources.

12. The device of claim **11**, wherein each of said light sources of said first light is a light emitting diode.

13. The device of claim **12**, wherein said second light comprises a plurality of light sources.

14. The device of claim **13**, wherein each of said light sources of said second light is a light emitting diode.

15. The device of claim **13**, wherein said third light comprises a plurality of light sources.

16. The device of claim **1**, wherein each of said light sources of said third light is a light emitting diode.

17. The device of claim **1**, wherein said inner bracket is rotatably connected to a stand.

* * * * *